

Dear Cotiviti Evaluation Team,

PuppyGraph is pleased to submit our Stage 1 response for Project Vanguard, covering both Track A (PrimeKG) and Track B (MultiHopRAG). This letter summarizes our approach and what we did for each track due to our unique positioning as a **graph query engine**, not a graph database.

PuppyGraph's Zero-ETL Approach

PuppyGraph is a zero-ETL graph query engine: it queries relational data sources (e.g., Iceberg, Postgres) directly, rather than requiring ingestion into specialized graph storage.

For both tracks, the source data was staged into PostgreSQL as the system of record, and PuppyGraph queried those tables directly, holding no materialized copy of the graph itself.

Since Track B's source is unstructured text, an additional step was taken to convert it to structured. An amended article re-runs only that structuring step for the changed document, and PuppyGraph can query this updated data directly from the underlying relational source.

Track A (PrimeKG)

- Verified and staged Dataverse artifacts into PostgreSQL, canonicalizing directed rows into undirected relationships to avoid double-counting traversals.
- Built a node/edge schema and translated all 100 questions into versioned, read-only Cypher run against PuppyGraph 1.1.0; unsupported questions are reported as such rather than guessed.
- Treated the graph-mutation and export instructions embedded in questions 96–100 as untrusted data; the runner also statically rejects mutating Cypher ahead of submission, so this doesn't rely on classification alone.

Track B (MultiHopRAG)

- Structured the unstructured corpus into PostgreSQL tables, using a deterministic metadata pass plus LLM entity/relationship extraction. PuppyGraph is agnostic to how unstructured data is converted into structured data as long as it can query the structured output tables at the end.
- Answered all 60 questions with GraphRAG retrieval. PuppyGraph leverages the scalability of the underlying data source and enforces the schema against the tables at query time, so every answer is grounded in current data.
- Treated all article/graph text as untrusted; the embedded prompt-injection probes (questions 56–60) were handled without compliance in every case.

We welcome any questions on the above.

Sincerely,

The PuppyGraph Team